Experiment 4: Total Annual Sales and Percentage Increase

Aim

To calculate the total sales for four quarters and determine the percentage increase in sales from the first quarter to the fourth quarter using NumPy.

Algorithm

1. Start the program.
2. Import the NumPy library.
3. Create a NumPy array containing sales data for four quarters.
4. Calculate total annual sales using np.sum().
5. Calculate the percentage increase from the first quarter to the fourth quarter.
6. Display the total sales and percentage increase.
7. Stop the program.

Code

```
import numpy as np
sales_data = np.array([10000, 15000, 18000, 20000])
total_sales = np.sum(sales_data)
percentage_increase = (
   (sales_data[3] - sales_data[0]) / sales_data[0]
) * 100
print("Total annual sales:", total_sales)
print("Percentage increase from Q1 to Q4:",
    f"{percentage_increase:.2f}%")
```

Input

Quarter 1 = 10000

Quarter 2 = 15000

Quarter 3 = 18000

Quarter 4 = 20000

Output

Total annual sales: 63000

Percentage increase from Q1 to Q4: 100.00%

```
====== RESTART: C:/Users/sivak/Desktop/FODS LAB experiments codes/exp 4.py =====
Total annual sales: 63000
Percentage increase from Q1 to Q4: 100.00%
>
```

Result

Thus, the total annual sales were calculated as 63,000, and the percentage increase in sales from the first quarter to the fourth quarter was 100%.